Doeun Kim

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Seoul, South Korea

EDUCATION

Sogang University Mar 2021 – Present
B.S. in Mathematics and Computer Science
GPA: **4.16 / 4.3 (4.38 / 4.5)** Expected graduation: **Feb 2027**

Leave of absence for mandatory military service (Oct 2022 – Apr 2024): KATUSA (Korean Augmentation to the U.S. Army).

RESEARCH INTERESTS

Machine Learning; Causal Machine Learning; Causal Foundation Models

RESEARCH EXPERIENCE

Causality Lab, SNU Graduate School of Data Science Jul 2025 – Present
Undergraduate Researcher (Advisor: Prof. Sanghack Lee) Seoul, South Korea

SKI-ML Lab, SNU Graduate School of Data Science Jun 2024 – Aug 2024
Intern (Advisor: Prof. Jay-Yoon Lee) Seoul, South Korea

WORKSHOP PAPERS

A Structural View of Query Misspecification in Causal Foundation Models
Junha Ham*, Deokgyu Kim*, **Doeun Kim***, Serjin Kim*, and Sanghack Lee†
ICML 2026 SPIGM Workshop — poster (non-archival). [Paper]

Causal Foundation Models Perform Better without Post-treatment Variables
Junha Ham*, Deokgyu Kim*, **Doeun Kim***, Serjin Kim*, and Sanghack Lee†
ICML 2026 FMSD Workshop — poster (non-archival). [Paper]

* Equal contribution. † Corresponding author.

PROJECTS

Julhaengrang: Subway Crowding Avoidance with ML & Pareto Optimization Summer 2026
- Built an ML-based station-avoidance service to address unavailable real-time crowding data, training station-specific models on five-minute observations across 7 stations to forecast passenger inflow and crowding probabilities.
- Combined ML forecasts with route estimates and Pareto optimization to compare 91 station–wait options by crowding, travel time, walking, and transfers.
- The project was **selected for further development** by KT and partner organizations.

Interactive 3D Graphics System with OpenGL & GLSL Spring 2026
- Developed an interactive 3D scene with multi-camera navigation, Catmull–Rom motion, animated-object tracking, and a seven-view CCTV interface using C++ and OpenGL.
- Implemented Phong lighting, texture mapping, alpha blending, dynamic objects, and custom GLSL effects including a holographic Menger cube and spiral image transition.

Statistical Modeling of NYC Airbnb Prices Spring 2026
- Analyzed NYC Airbnb listings in SAS, engineering price, availability, property, and host features and applying hypothesis tests, categorical data analysis, ANOVA, and regression diagnostics.
- Built and compared multiple-regression and variable-selection models on 19,995 cases; the full model explained 41.59% of log-price variance and identified guest capacity, borough, and room type as the strongest predictors.

Do-PFN Really Do Causality? Fall 2025
- Collaborated with Causality Lab interns on a research project examining whether Do-PFN/PFN-style models exhibit causal reasoning behavior.

FA-AL-MOT: Multimodal Fashion Price Prediction Fall 2025

- Built a multimodal AI model for C2C fashion resale price prediction and product attribute classification using image and text features.
- Used CLIP fusion and LLM-based attribute restoration.

DIGIRO: Real-Time Notification Service Summer 2025

- Architected and developed a real-time notification service using FastAPI (Python) and PostgreSQL to aggregate and serve live data for Seoul's Jongno-gu district.
- Containerized the application's backend, crawler, and database services using Docker and managed deployment orchestration with Kubernetes for scalability and reliability.
- Integrated multiple external sources, including Seoul Public Data and Kakao Maps APIs, to provide users with an interactive map visualizing protest locations, routes, and traffic control areas.

News Summarization with GPT-2 & Knowledge Distillation Spring 2025

- Enhanced GPT-2 summarization on CNN/DailyMail dataset; sequence-level distillation from LLaMA3-8B improved ROUGE-1 by 20.6% and METEOR by 54.6%.
- Analyzed decoder-only model limitations and cross-model distillation challenges.

Convenience-Store Management Application Spring 2025

- C/MySQL CLI tool for business analytics queries.

Automated Lighting Control via Object Detection Winter 2024

- Integrated YOLO + ByteTrack with a line-crossing counter to trigger lights based on occupancy.
- Received the **Excellence Award** in Sogang University's Inquiry Community Program (Jan 2026).

RAG Chatbot for Academic Catalog Fall 2024

- Built a RAG pipeline with SOLAR embeddings, SOLAR API, ChromaDB, and Streamlit UI for real-time Q&A on Academic Catalog.

Fine-Grained Classification Spring 2024

- Developed ResNet-101 classifier on CUB-200-2011; augmented data with Gaussian noise to defend against PGD attacks.

HONORS & SCHOLARSHIPS

Outstanding Scholar, KT; one of four recipients Jan 2026

Excellence Award, Sogang University, Inquiry Community Program, for the automated lighting control project Jan 2026

KT Digital Talent Scholarship, KT (Full tuition for four semesters) Spring 2025 – Fall 2026

Certificate of Appreciation (COA), awarded by COL David F. Henning, Garrison Commander, U.S. Army Garrison Daegu 16 Apr 2024

Dean's List, Sogang University, College of Natural Sciences (2021 academic year) 2022

President's Award for Educational Excellence, President of the United States 2015

ACADEMIC SERVICE

Volunteer Note-Taker, Computer Networks(CSE4175) Spring 2026
Sogang University

- Prepared lecture notes to support course accessibility for a student with a hearing impairment.

Teaching Assistant, Introduction to AI Programming(COR1010) Winter 2024; Summer 2025
Sogang University

- Answered programming questions, assisted students with assignments, and graded coursework.

TECHNICAL SKILLS & LANGUAGES

Programming & data: Python, C/C++, SQL, SAS, MATLAB

Tools: Git, PostgreSQL/MySQL, LaTeX

Languages: Korean (native), English (TEPS 506/600)